

FCAI ANU Materials – Semi Seniors – Chapter 4 – Computer Networks

Multiple Choice Questions (120)

4.1 Overview of Network: Data and Control Planes

1. The network layer can be decomposed into two interacting parts, the data plane and the control plane. Which of the following describes the function of the **data plane**?

- A. Coordinating local, per-router forwarding actions
- B. Determining the end-to-end path packets take from source to destination
- C. Per-router functions that determine how a datagram arriving on one input link is forwarded to an output link
- D. Calculating the route or path taken by packets

Answer: C

2. What is the primary data-plane role of each router?

- A. Running routing protocols
- B. Forwarding datagrams from its input links to its output links
- C. Computing the forwarding table
- D. Determining the end-to-end paths packets take

Answer: B

3. What is the primary role of the network **control plane**?

- A. Transferring a packet from an input link interface to the appropriate output link interface
- B. Coordinating local, per-router forwarding actions so that datagrams are ultimately transferred end-to-end
- C. Transferring packets from a sending host to a receiving host
- D. Executing the lookup function at the input port

Answer: B

4. Forwarding is defined as the router-local action of transferring a packet from an input link interface to the appropriate output link interface. How quickly does this action typically take place?

- A. Typically seconds
- B. Typically milliseconds
- C. Typically a few nanoseconds
- D. Typically minutes

Answer: C

5. Which key network-layer function refers to the network-wide process that determines the end-to-end paths that packets take from source to destination?

- A. Forwarding
- B. Encapsulation
- C. Routing

D. Data plane processing

Answer: C

6. Routing algorithms calculate the paths packets take and are implemented in which plane of the network layer?

- A. Data plane
- B. Forwarding plane
- C. Control plane
- D. Transport plane

Answer: C

7. A key element in every network router used for the forwarding function is its:

- A. Routing protocol exchange table
- B. Control processor
- C. Routing algorithm
- D. Forwarding table

Answer: D

8. In traditional IP forwarding, how is the forwarding decision primarily based?

- A. Using values in several different fields in the datagram's header
- B. Based on a datagram's destination address
- C. Based on the Time-to-Live field
- D. Based on the source IP address

Answer: B

9. Generalized forwarding involves functions performed using values in which part of the packet?

- A. Only the destination address field
- B. Only the source address field
- C. Several different fields in the datagram's header
- D. Only the Protocol field

Answer: C

10. The process where a packet might be blocked from exiting a router, or duplicated and sent over multiple outgoing links, falls under which function?

- A. Routing
- B. Forwarding
- C. Generalized forwarding
- D. Control plane actions

Answer: B

11. In the traditional approach to the control plane, where does the routing algorithm run?

- A. In a physically separate, remote controller
- B. In the end host systems (H1 and H2)
- C. In each and every router

D. Only in the destination router (R2)

Answer: C

12. How does the routing algorithm in one router communicate with others to compute forwarding table values?

- A. Through manual human configuration
- B. By exchanging routing messages according to a routing protocol
- C. Via the switching fabric
- D. Through the transport layer

Answer: B

13. The control-plane approach with a physically separate, remote controller is central to which concept?

- A. Traditional IP Forwarding
- B. Network Address Translation (NAT)
- C. Dynamic Host Configuration Protocol (DHCP)
- D. Software-Defined Networking (SDN)

Answer: D

14. What happens to control-plane routing functionality in the SDN approach?

- A. It remains within the physical router
- B. It is separated from the physical router, handled by a remote controller
- C. It is eliminated entirely
- D. It is implemented entirely in hardware

Answer: B

15. What typically defines the characteristics of end-to-end delivery of packets between sending and receiving hosts?

- A. The transport-layer protocol
- B. The network service model
- C. The routing algorithm
- D. The link-layer protocol

Answer: B

16. Which of the following is NOT a possible network layer service model?

- A. Guaranteed delivery with bounded delay
- B. Guaranteed minimal bandwidth
- C. Guaranteed minimum packet size
- D. In-order packet delivery

Answer: C

17. The Internet's network layer provides a single service model known as:

- A. Guaranteed delivery
- B. In-order delivery
- C. Best-effort service

D. Guaranteed minimal bandwidth

Answer: C

18. Under best-effort service, which guarantee is provided regarding packet delivery or order?

- A. Packets are guaranteed to arrive eventually
- B. Packets are guaranteed to arrive in the order sent
- C. Packets are guaranteed delivery within a specified delay bound
- D. None of the above; delivery is not guaranteed, nor is order or delay

Answer: D

19. Why is forwarding typically implemented in hardware?

- A. Because it is a network-wide process
- B. Because it takes place on longer timescales (seconds)
- C. Because it takes place at very short timescales (typically a few nanoseconds)
- D. Because it involves complex routing protocol exchanges

Answer: C

20. A packet arriving from Host H1 to Router R1 must be forwarded to the next router on a path to H2. This action is an example of:

- A. Routing
- B. Forwarding
- C. Encapsulation
- D. Datagram extraction

Answer: B

21. A generic router architecture is typically comprised of four main components. Which component performs physical and link-layer termination functions, along with the crucial lookup function?

- A. Switching fabric
- B. Output ports
- C. Routing processor
- D. Input ports

Answer: D

22. Which component of a router is responsible for executing routing protocols, maintaining routing tables, and computing the forwarding table in traditional routers?

- A. Switching fabric
- B. Input ports
- C. Output ports
- D. Routing processor

Answer: D

23. Which component connects the router's input ports to its output ports and is completely contained within the router?

- A. Routing processor
- B. Switching fabric

- C. Line card
- D. Physical link termination

Answer: B

24. In SDN routers, the routing processor communicates with the remote controller primarily to:

- A. Perform the lookup function directly on every packet
- B. Transmit packets on the outgoing link
- C. Receive forwarding table entries and install them in the input ports
- D. Handle the physical layer functions

Answer: C

25. While the data plane operates at the nanosecond time scale, control plane functions like executing routing protocols typically operate at which time scale?

- A. Picosecond
- B. Millisecond or second
- C. Microsecond
- D. Constant time (zero delay)

Answer: B

26. Why is the forwarding table copied from the routing processor to the line cards (input ports)?

- A. To allow the routing processor to manage network management functions
- B. To ensure that routing algorithms run in every router
- C. To avoid a centralized processing bottleneck by allowing forwarding decisions to be made locally on a per-packet basis
- D. To facilitate the encapsulation of segments into datagrams

Answer: C

27. If forwarding decisions were based on a brute-force implementation of the forwarding table using 32-bit IP addresses, approximately how many entries would be required?

- A. 32 entries
- B. 2^{16} (65,536) entries
- C. More than 4 billion entries
- D. 256 entries

Answer: C

28. When a destination address matches multiple entries in a router's forwarding table, which rule is used to determine the correct output link?

- A. First-match rule
- B. Longest prefix matching rule
- C. Shortest prefix matching rule
- D. Default interface rule

Answer: B

29. Longest prefix matching is critical because lookups must be performed very quickly. At Gigabit transmission rates, how fast must this lookup be?

- A. In seconds

- B. In milliseconds
- C. In microseconds
- D. In nanoseconds

Answer: D

30. Which abstraction, performed in many networked devices including routers, firewalls, and NAT devices, involves determining the output port based on header fields and executing a specific operation?

- A. Encapsulation and Decapsulation
- B. Lookup and Queueing
- C. Match plus action
- D. TTL decrementing

Answer: C

4.2 What's Inside a Router?

31. In Network Address Translation (NAT), what action might be taken for an incoming packet whose transport-layer port number matches a given value?

- A. The packet is duplicated
- B. The packet is dropped (action)
- C. Its port number is rewritten before forwarding (action)
- D. The packet is stored in the output port's memory

Answer: C

32. Which of the following mandatory actions must occur during input port processing, besides the lookup function?

- A. Running the entire routing protocol suite
- B. Physical- and link-layer processing
- C. Reassembling fragmented datagrams
- D. Computing the overall end-to-end path

Answer: B

33. What must be checked and potentially rewritten for an IP packet during input port processing, as related to network management or loop prevention?

- A. Source IP address and Destination IP address
- B. Version number, checksum, and time-to-live (TTL) field
- C. Traffic class and Flow label
- D. Payload length

Answer: B

34. In the switching via memory approach, how does an input port initially signal the routing processor about an arriving packet?

- A. Via a separate bus connection
- B. Via an interrupt
- C. Via the switching fabric

D. Via the output port

Answer: B

35. For a router switching via memory, if the memory bandwidth supports a maximum of B packets per second read/write rate, what is the maximum overall forwarding throughput?

- A. B packets per second
- B. 2B packets per second
- C. Less than B/2 packets per second
- D. $N \times R_{line}$ packets per second

Answer: C

36. In switching via memory for early routers, why couldn't two packets be forwarded at the same time, even if they had different destination ports?

- A. Because the routing processor could only handle one interrupt at a time
- B. Because only one memory read/write can be done at a time over the shared system bus
- C. Because of Head-of-the-Line (HOL) blocking
- D. Because the input port needed to pre-pend a label

Answer: B

37. In the switching via a bus approach, how does an input port indicate the local output port to which the packet should be transferred?

- A. By setting the destination IP address to the output port's IP
- B. By transmitting the packet over a crossbar switch
- C. By pre-pending a switch-internal label (header) to the packet
- D. By signaling the centralized routing processor

Answer: C

38. What is the main limitation of switching via a single, shared bus?

- A. It requires intervention by the routing processor
- B. It is limited to the bus speed, as every packet must cross the single bus
- C. It leads to HOL blocking
- D. It can only support small numbers of ports

Answer: B

39. A crossbar switch is an interconnection network consisting of how many buses connecting N input ports to N output ports?

- A. N buses
- B. 2N buses
- C. N^2 buses
- D. 32 buses

Answer: B

40. What is a key advantage of a crossbar switch over switching via memory or a single bus?

- A. It is less expensive to implement
- B. It eliminates all forms of queuing
- C. It is capable of forwarding multiple packets in parallel

D. It uses the routing processor for every forwarding decision

Answer: C

41. A crossbar switch is defined as non-blocking. What does this mean?

- A. A packet will never be blocked regardless of other traffic
- B. A packet being forwarded will not be blocked from reaching its output port as long as no other packet is currently being forwarded to that output port
- C. All packets are forwarded using the longest prefix match rule
- D. No queuing occurs at the input ports

Answer: B

42. When does queuing occur at a router?

- A. Only at the input ports
- B. Only at the output ports
- C. At both the input ports and the output ports
- D. Only if the switching fabric is non-blocking

Answer: C

43. If a router has N input ports and N output ports with identical line speed R_{line} , what switching fabric transfer rate (R_{switch}) is required for only negligible queuing to occur at the input ports?

- A. $R_{switch} = R_{line}$
- B. $R_{switch} = N \times R_{line}$
- C. R_{switch} must be infinitely fast
- D. $R_{switch} = R_{line} / N$

Answer: B

44. The phenomenon where a queued packet in an input queue must wait for transfer through the switching fabric even though its output port is free, because it is blocked by another packet at the front of the line, is known as:

- A. Output contention
- B. Drop-tail policy
- C. Head-of-the-Line (HOL) blocking
- D. Crossbar saturation

Answer: C

45. If the switching fabric is N times faster than the line speeds ($R_{switch} = N \times R_{line}$), and all N input packets are destined for the same output port, where will packet queues form?

- A. Input ports only
- B. Only at the routing processor
- C. Output ports only
- D. Neither input nor output queues form

Answer: C

46. When the output port memory is full and a decision must be made to drop the arriving packet, this policy is known as:

- A. Active Queue Management (AQM)
- B. Random Early Detection (RED)
- C. Drop-tail
- D. Head-of-the-Line blocking

Answer: C

47. What is the general name for proactive packet-dropping and -marking policies proposed to provide a congestion signal to the sender?

- A. Longest Prefix Matching
- B. Weighted Fair Queuing (WFQ)
- C. Active Queue Management (AQM) algorithms
- D. First-In-First-Out (FIFO)

Answer: C

48. Which of the following is listed as one of the most widely studied and implemented Active Queue Management (AQM) algorithms?

- A. FIFO
- B. Random Early Detection (RED)
- C. Proportional Integral controller Enhanced (PIE)
- D. Weighted Fair Queuing (WFQ)

Answer: B

49. For many years, the rule of thumb for buffer sizing (B) was calculated based on which formula?

- A. $B = RTT \times C$
- B. $B = N \times RTT \times C$
- C. $B = N / R_{line}$
- D. B = 4 billion entries

Answer: A

50. More recent theoretical and experimental efforts suggest that when a large number of independent TCP flows (N) pass through a link, the buffering needed (B) is proportional to:

- A. $B = RTT \times C$
- B. $B = N \times RTT \times C$
- C. $B = N$
- D. $B = C$

Answer: B

51. Larger router buffers are considered a double-edged sword because while they absorb short-term statistical traffic fluctuations, they can also lead to:

- A. Increased packet loss rates
- B. Decreased memory usage
- C. Increased delay
- D. More complicated hardware implementation

Answer: C

52. What is another name for the First-Come-First-Served (FCFS) link-scheduling discipline?

- A. Priority Queuing
- B. Weighted Fair Queuing (WFQ)
- C. First-In-First-Out (FIFO)
- D. Round Robin

Answer: C

53. In a FIFO queue, if packet 1 arrives at $t = 0$ and takes 3 units of time to transmit, and packet 3 arrives at $t = 4$, when does packet 3 begin service?

- A. $t = 0$
- B. $t = 3$
- C. $t = 6$
- D. $t = 9$

Answer: C

54. Under priority queuing, when choosing a packet to transmit, the discipline will transmit a packet from which queue?

- A. The queue with the longest waiting time
- B. The queue that arrived first chronologically
- C. The highest priority class that has a nonempty queue
- D. The lowest priority class

Answer: C

55. In priority queuing, how is the choice typically made among packets within the same priority class?

- A. Randomly
- B. In a Round Robin manner
- C. In a FIFO manner
- D. Based on packet size

Answer: C

56. In non-preemptive priority queuing, if a low-priority packet is transmitting and a high-priority packet arrives, when does the high-priority packet begin transmission?

- A. Immediately
- B. After the current packet completes
- C. Never
- D. After the next packet departs

Answer: B

57. Under the round robin queuing discipline, how is service alternated among packet classes?

- A. Strict priority is enforced
- B. Service is strictly FCFS
- C. Service alternates in turn among the classes
- D. Only the highest priority class is served

Answer: C

58. What is the definition of a work-conserving queuing discipline?

- A. It prioritizes high-priority traffic
- B. It alternates service among classes
- C. It guarantees delivery within a bounded delay
- D. It will never allow the link to remain idle whenever there are packets queued for transmission

Answer: D

59. What generalized form of round robin queuing has been widely implemented in routers?

- A. FCFS
- B. Random Early Detection (RED)
- C. Weighted Fair Queuing (WFQ)
- D. Non-preemptive priority queuing

Answer: C

60. In two-class round robin (Class 1: Packets 1,2,4; Class 2: Packets 3,5), Packet 1 is transmitted first. Which packet is transmitted next?

- A. Packet 2
- B. Packet 4
- C. Packet 3
- D. Packet 5

Answer: C

4.3 The Internet Protocol (IP)

61. What is the Internet's network-layer packet referred to as?

- A. Segment
- B. Frame
- C. Datagram
- D. Cell

Answer: C

62. In the IPv4 datagram format, how many bits specify the IP protocol version?

- A. 16 bits
- B. 32 bits
- C. 4 bits
- D. 8 bits

Answer: C

63. Why is the Header Length field necessary in the IPv4 datagram?

- A. To determine the maximum payload size
- B. Because an IPv4 datagram can contain a variable number of options
- C. To specify the protocol of the data portion
- D. To detect bit errors

Answer: B

64. What is the typical size of an IPv4 datagram header when no options are included?

- A. 40 bytes
- B. 20 bytes
- C. 16 bits
- D. 65,535 bytes

Answer: B

65. Since the Datagram Length field in IPv4 is 16 bits long, what is the theoretical maximum size of an IP datagram, measured in bytes?

- A. 1,500 bytes
- B. 20 bytes
- C. 65,536 bytes
- D. 65,535 bytes

Answer: D

66. What is the purpose of the Time-to-Live (TTL) field in the IPv4 datagram?

- A. To ensure datagrams do not circulate forever in the network
- B. To specify the maximum delay allowed
- C. To indicate the total length of the datagram
- D. To coordinate with the transport layer

Answer: A

67. If the TTL field reaches 0, what action must a router take?

- A. Forward the packet using the default interface
- B. Forward the packet to the routing processor
- C. Drop that datagram
- D. Recompute the TTL checksum

Answer: C

68. The Protocol field in the IPv4 datagram, when it reaches the final destination, indicates what?

- A. The specific link-layer protocol to use
- B. The specific application layer protocol
- C. The specific transport-layer protocol to which the data portion should be passed
- D. The router's priority class

Answer: C

69. If the Protocol field in an IPv4 datagram has a value of 6, to which transport-layer protocol is the data portion passed?

- A. UDP
- B. ICMP
- C. TCP
- D. IPv6

Answer: C

70. Why must the Header Checksum be recomputed and stored again at each router that processes an IPv4 datagram?

- A. Because TCP/IP performs error checking at both the transport and network layers
- B. Because the Header Length field changes
- C. Because the TTL field (and possibly options field) will change
- D. Because the Source IP address is rewritten

Answer: C

71. An IP address is technically associated with what?

- A. The host or router itself
- B. The transport layer segment
- C. An interface on the host or router
- D. The switching fabric

Answer: C

72. How long is each IP address (IPv4)?

- A. 128 bits
- B. 4 bytes or 32 bits
- C. 16 bits
- D. 20 bytes

Answer: B

73. IPv4 addresses are typically written in what notation?

- A. Binary notation
- B. Hexadecimal notation
- C. Dotted-decimal notation
- D. CIDR notation

Answer: C

74. In IP terms, a routerless network connecting multiple host interfaces and one router interface forms a(n):

- A. Interconnection Network
- B. Subnet
- C. ICANN
- D. Control Plane

Answer: B

75. What does the "/24" notation in the subnet address 223.1.1.0/24 indicate?

- A. The length of the address in bytes
- B. The number of hosts attached to the subnet
- C. That the leftmost 24 bits of the 32-bit quantity define the subnet address
- D. The number of router interfaces present

Answer: C

76. The Internet's current address assignment strategy that generalizes the notion of subnet addressing is known as:

- A. Classful Addressing
- B. Static Addressing

- C. Classless Interdomain Routing (CIDR)
- D. Network Address Translation (NAT)

Answer: C

77. Under CIDR, the x most significant bits of an address a.b.c.d/x constitute the network portion and are referred to as the:

- A. Host identifier
- B. Prefix (or network prefix)
- C. Subnet mask length
- D. Flow label

Answer: B

78. Why does CIDR reduce the size of the forwarding table in routers outside an organization's network?

- A. Because all 32 bits are considered for forwarding
- B. Because only the x leading prefix bits need to be considered
- C. Because routing is implemented in hardware
- D. Because the organization uses NAT

Answer: B

79. Before CIDR was adopted, addressing was constrained to network portions of 8, 16, or 24 bits in length. What was this scheme known as?

- A. Prefix Matching
- B. Flexible Addressing
- C. Classful Addressing
- D. Interdomain Routing

Answer: C

80. A Class C (/24) subnet under classful addressing could accommodate how many hosts?

- A. 65,536 hosts
- B. Up to 254 hosts
- C. 2^{32} hosts
- D. 2^{24} hosts

Answer: B

81. What is the IP broadcast address used to deliver a datagram to all hosts on the same subnet?

- A. 0.0.0.0
- B. 127.0.0.1
- C. 255.255.255.255
- D. 10.0.0.0

Answer: C

82. Which organization has the ultimate responsibility for managing the IP address space, allocating address blocks, and managing the DNS root servers?

- A. IETF
- B. IEEE

- C. ICANN
- D. ISP

Answer: C

83. Which protocol allows a host to obtain an IP address and other configuration information automatically?

- A. NAT
- B. BGP
- C. DHCP
- D. TCP

Answer: C

84. In addition to the host IP address assignment, what information can DHCP allow a host to learn?

- A. The source IP address of the destination host
- B. The address of its first-hop router (default gateway)
- C. The optimal path using the routing algorithm
- D. The amount of buffering required at the router

Answer: B

85. DHCP is described as a client-server protocol. What is typically the role of a client?

- A. A newly arriving host wanting to obtain network configuration information
- B. The remote controller in an SDN architecture
- C. The input port processing unit
- D. The network administrator

Answer: A

86. If no DHCP server is present on a subnet, what device might be needed to know the address of a DHCP server for that network?

- A. A crossbar switch
- B. A DHCP relay agent (typically a router)
- C. The destination host (H2)
- D. A layer 3 firewall

Answer: B

87. What is the first step in the DHCP protocol process for a newly arriving host?

- A. DHCP Request
- B. DHCP ACK
- C. DHCP Server Discovery
- D. DHCP Offer

Answer: C

88. When sending a DHCP discover message, what specific source IP address does the client use?

- A. Its own IP address
- B. The broadcast address (255.255.255.255)

- C. The address of the DHCP server
- D. A "this host" source IP address of 0.0.0.0

Answer: D

89. To ensure the DHCP discover message reaches all nodes on the subnet, what destination IP address is used?

- A. 0.0.0.0
- B. 127.0.0.1
- C. 255.255.255.255 (broadcast)
- D. The router's interface address

Answer: C

90. In the DHCP four-step process, which message contains the proposed IP address for the client and the IP address lease time?

- A. DHCP Discover
- B. DHCP Offer
- C. DHCP Request
- D. DHCP ACK

Answer: B

4.3 The Internet Protocol (IP) – Continued

91. What technique is used to solve the problem of IPv4 address depletion by allowing a small office/home office (SOHO) network with multiple devices to use a single globally unique IP address?

- A. CIDR
- B. Classful Addressing
- C. DHCP
- D. Network Address Translation (NAT)

Answer: D

92. In a NAT-enabled router scenario, what address range is typically used internally within the home network for local devices?

- A. Globally unique, ISP-provided addresses
- B. An address block like 10.0.0.0/24
- C. The WAN-side IP address
- D. Broadcast addresses

Answer: B

93. To the outside world (WAN), how does a NAT-enabled router appear?

- A. As multiple devices with unique IP addresses
- B. As a single device with a single IP address
- C. As a DHCP server
- D. As a remote controller

Answer: B

94. If all datagrams arriving at the NAT router from the WAN have the same destination IP address (the router's WAN-side address), how does the router determine the internal host to forward a given datagram?

- A. By examining the TTL field
- B. By using a NAT translation table involving port numbers and IP addresses
- C. By consulting the routing processor
- D. By applying the longest prefix matching rule

Answer: B

95. In the NAT translation table example, when the internal host 10.0.0.1 (port 3345) sends a request, the NAT router rewrites the source port to 5001. When the server responds, which destination port number must the server use?

- A. The original source port 3345
- B. The rewritten port 5001
- C. The well-known port 80
- D. The broadcast port 67

Answer: B

96. How many simultaneous connections can the NAT protocol support with a single WAN-side IP address, given that the port number field is 16 bits long?

- A. Approximately 4 billion
- B. Exactly 65,535
- C. Over 60,000
- D. 254

Answer: C

97. What is a key philosophical argument raised against NAT by architectural purists?

- A. It is too difficult to implement in hardware
- B. It violates the principle that routers, as layer 3 devices, should not modify IP addresses or look at port numbers
- C. It relies on DHCP, which is not robust
- D. It requires classful addressing

Answer: B

98. What was the prime motivation for the IETF to develop IPv6 in the early 1990s?

- A. To standardize QoS requirements
- B. IPv4 address space exhaustion
- C. To simplify router architecture
- D. To eliminate NAT traversal issues

Answer: B

99. What is the size of the IP address in IPv6?

- A. 32 bits
- B. 40 bytes
- C. 128 bits

D. 64 bits

Answer: C

100. What is the fixed length of the streamlined header in the IPv6 datagram?

A. 20 bytes

B. 40 bytes

C. 60 bytes

D. 128 bits

Answer: B

101. What value does the Version field in an IPv6 datagram carry?

A. 4

B. 6

C. 32

D. 128

Answer: B

102. Which IPv6 field allows labeling of packets belonging to particular flows for which the sender requests special handling?

A. Traffic class

B. Payload length

C. Flow label

D. Next hdr

Answer: C

103. In IPv6, the Payload Length field gives the number of bytes in the datagram following the fixed-length header. How long is this field?

A. 4 bits

B. 8 bits

C. 16 bits

D. 128 bits

Answer: C

104. What is the IPv6 equivalent of the IPv4 Time-to-Live (TTL) field?

A. Traffic class

B. Next hdr

C. Hop limit

D. Flow label

Answer: C

105. Which of the following operations is explicitly not allowed at intermediate routers in IPv6?

A. Forwarding the datagram

B. Decrementing the Hop limit

C. Fragmentation and reassembly

D. Lookup using the destination address

Answer: C

106. If an IPv6 datagram is too large to be forwarded over an outgoing link, what action does the router take?

- A. It fragments the datagram
- B. It drops the datagram and sends a "Packet Too Big" ICMP error
- C. It queues the datagram at the input port
- D. It rewrites the Payload Length field

Answer: B

107. What IPv4 header field was removed from IPv6 because transport-layer and link-layer protocols already provide this functionality?

- A. Header length
- B. Source and destination addresses
- C. Protocol field
- D. Header checksum

Answer: D

108. The removal of the Header Checksum in IPv6 contributes to faster IP forwarding because:

- A. Checksum calculation is now performed in software
- B. Checksumming is only performed at the destination host
- C. In IPv4, checksum needed to be recomputed at every router (TTL changes)
- D. The IPv6 header is 128 bits long

Answer: C

109. Why were options not included as part of the standard IPv6 header?

- A. They were irrelevant in IPv6
- B. IPv4 options complicated processing due to variable header length
- C. Options are handled entirely by the routing processor
- D. Options are handled by the Flow label field

Answer: B

110. The transition from IPv4 to IPv6 rules out a "flag day." What does the "flag day" concept involve?

- A. Manually configuring all forwarding tables
- B. Allowing routers to handle both IPv4 and IPv6 simultaneously
- C. A specific time when all Internet machines are upgraded from IPv4 to IPv6
- D. Using NAT to manage addresses

Answer: C

111. What is the basic idea behind tunneling as a mechanism for IPv4-to-IPv6 transition?

- A. Implementing IPv6 over a crossbar switch
- B. Using DHCP to assign IPv6 addresses
- C. Encapsulating the IPv6 datagram inside an IPv4 datagram
- D. Running the IPv6 routing protocol over IPv4

Answer: C

112. In tunneling, what type of datagram do intervening IPv4 routers route?

- A. IPv6 datagrams
- B. Encapsulated transport segments
- C. IPv4 datagrams containing IPv6 payload
- D. DHCP messages

Answer: C

113. In a tunnel from IPv6 Node B to Node E, what is the destination IP of the IPv4 datagram created by B?

- A. Ultimate destination
- B. Next router
- C. Receiving IPv6 node E
- D. Broadcast address

Answer: C

114. One important lesson from the IPv6 experience regarding network layer protocols is:

- A. Changing network-layer protocols is easy
- B. Introducing new network-layer protocols is difficult and slow
- C. Network-layer changes occur faster than application-layer
- D. NAT solves depletion entirely

Answer: B

115. Introducing new protocols at the application layer is analogous to:

- A. Replacing the foundation of a house
- B. Tearing the house down
- C. Adding a new layer of paint
- D. Running algorithms in the basement

Answer: C

116. When H1 sends information to H2, the network layer in H1:

- A. Extracts transport segments
- B. Forwards datagrams to output link
- C. Encapsulates each segment into a datagram
- D. Determines end-to-end route

Answer: C

117. How is routing distinguished from forwarding in terms of implementation location?

- A. Forwarding in software; routing in hardware
- B. Forwarding in hardware; routing often in software
- C. Both in switching fabric
- D. Both in input port

Answer: B

118. In a link-layer switch, what determines the action taken under "Match plus Action"?

- A. Destination IP addresses
- B. Time-to-live values

- C. Link-layer destination addresses
- D. Transport-layer port numbers

Answer: C

119. If a router uses a crossbar switch and two packets from different inputs target the same output, what happens to one packet?

- A. Both forwarded simultaneously
- B. Both dropped
- C. One waits at input
- D. Longest prefix matching rule applies

Answer: C

120. In IPv6, the Next hdr field serves a similar function to which IPv4 field?

- A. Header Length
- B. Datagram length
- C. Protocol
- D. Fragmentation offset

Answer: C

Wish you all the best!

Fathy.